

CrewBrief

The crew's pre-job briefing, turned into a completed hazard assessment

Voice AI for utility field crews · Built on Azure

Every job starts with a 40-minute form

40

minutes to complete a hazard assessment (JSA / SWMS) by hand, before every job



Time out of productive work

Crews fill the form in on the tailgate while the job waits.



Quality depends on who holds the pen

Hazards discussed out loud never make it onto paper.



Missed controls stay missed

Nobody checks that a critical hazard has a real control until an audit.

Crews already say it out loud. Capture it.

Today

- Crew talks through the job
- Leading hand writes the form afterwards
- Details are lost between the two

With CrewBrief

- Crew talks through the job, as normal
- The form drafts itself from what was said
- The crew lead reviews and signs in minutes

How it works: one standup, one signed form



1 Record

One device records the whole crew's briefing. Live, or upload a recording.



2 Understand

Azure AI Speech separates the voices. AI maps what was said to the company hazard library.



3 Clarify

No interruptions. At the end, CrewBrief asks only the critical questions it still has.



4 Sign

The crew lead reviews the drafted form with evidence for every field, then signs and submits.



Live demo

LV service connection · Pole replacement with a crane · Night storm restoration

Built so a wrong form never looks complete

Evidence for every field

Each hazard and control links to the exact words the crew said.

Safety rules in code, not prompts

A critical hazard needs an isolation or engineering control. Absence is never assumed.

A person always signs

The form is never submitted without the crew lead's sign-off.

Full audit trail

Every change records its source, model and prompt version, and who approved it.

Measured, not assumed

100%

Critical hazard recall

Target ≥ 98%

100%

Fields backed by evidence

Target ≥ 95%

0%

Unsupported (hallucinated) fields

Target ≤ 2%

Offline evaluation on 6 synthetic briefings (sample set). Every pull request runs this gate in CI. Next: 20 real briefings scored by safety experts, then a shadow pilot.

Enterprise-ready on Azure

Azure AI Speech

Transcription and speaker separation

Azure OpenAI

Understands the briefing, checks the draft

Azure AI Search

Company hazard and control library

Cosmos DB

Drafts, transcripts, audit log

Web PubSub

Live updates to the crew's screen

Container Apps

Autoscaling services for 1,000+ crews

Deterministic orchestrator (not an LLM) decides what runs when, so every decision is auditable. Infrastructure as code (Bicep, azd); CI with an accuracy gate.

Privacy and security by default



No raw audio stored

Transcripts only, with personal details redacted.



No voice biometrics

Voices are separated, never identified by voiceprint.



Crews are told

A recording notice is acknowledged before every briefing.



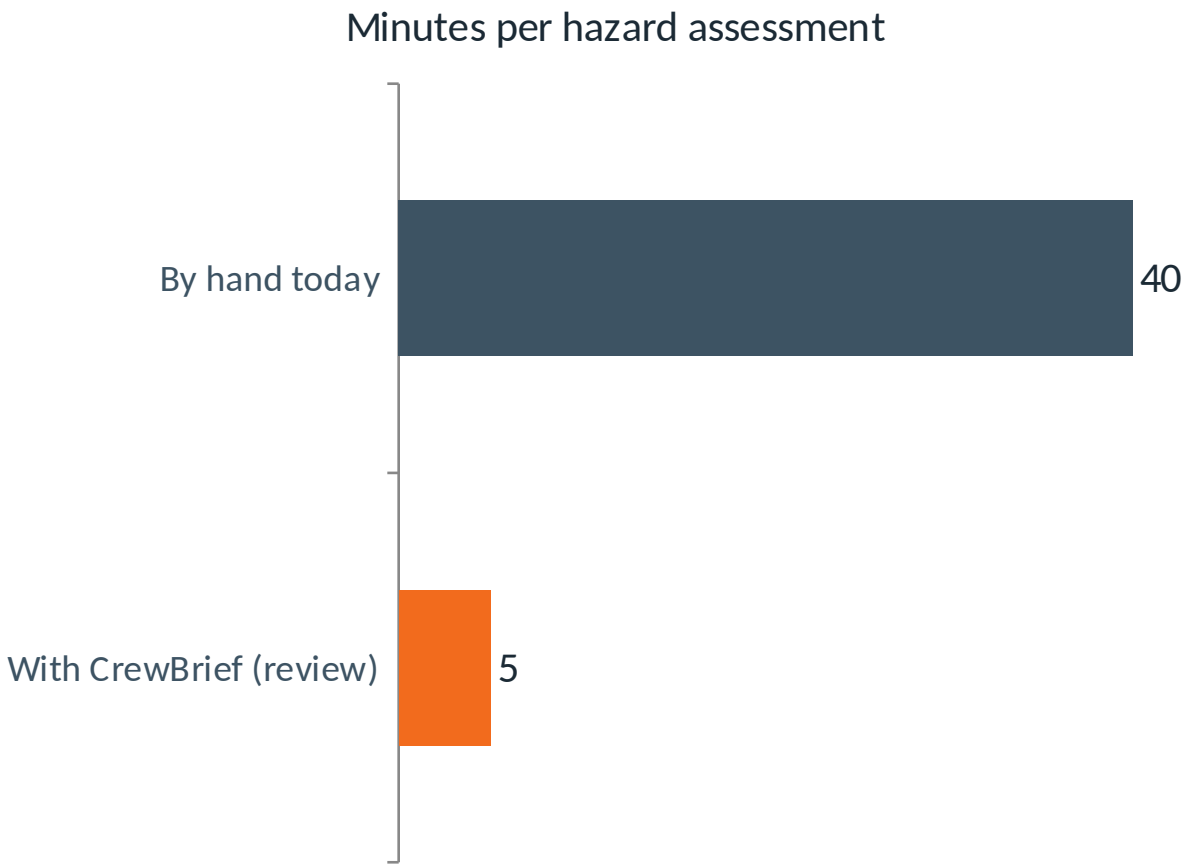
Australian data residency

Private networking, managed identity, Entra ID sign-in.

Deployment needs legal review under workplace surveillance and listening-device laws in each state.

CrewBrief assists with hazard assessment. It does not replace the crew's responsibility for job safety.

The time comes back to the job



≈580

crew-hours back every working day across 1,000 crews (35 minutes saved per form, one form per crew per day)

Illustrative estimate

From prototype to every crew



Prototype

Form schema, hazard library, live pipeline, standup clarifications, eval gate

Gold data

20 real briefings scored by safety experts; Azure speech tuned on utility terms

Shadow pilot

10 crews fill forms as today while CrewBrief drafts in parallel

Rollout

Provisioned throughput, multi-region speech, offline mode for poor coverage

Later: work-order and switching-instruction context, weather and fire-danger inputs.



Let's pilot it with 10 crews

We need: a pilot depot, 20 recorded briefings for the gold set, and an Azure subscription in Australia East